Abhishek Jha

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Education

New York University (Courant Institute) — New York, NY
Master of Science in Computer Science — *Sep 2025 – May 2027*
Relevant coursework: Operating Systems, Deep Learning, Machine Learning, Fundamental Algorithms

Delhi Technological University — New Delhi, India
Bachelor of Technology in Mechanical Engineering; GPA: 4.00/4.00 — *Dec 2020 – May 2024*
Relevant coursework: Machine Learning, Computer Vision, Optimization Techniques

Work Experience

NYU Langone Health — New York, NY
Student Intern — *Jan 2026 – Present*

- Working on an exam-level pipeline that aggregates multi-view mammograms into a single risk estimate across 1–5 year horizons, enforcing strict label hygiene to avoid leakage.
- Developing the domain-level self-supervised architecture and object detection with registration stack for FFDM, applying mammography-safe transforms, and using stratified sampling to balance positives/negatives while preserving data distributions.
- Running evaluation and analysis by computing horizon-wise and time-dependent AUC at 5 years determining risk of cancer, investigating horizon gaps, and performing targeted error analysis.

Magicpin — New Delhi, India
Associate Analyst Intern — *Aug 2024 – Dec 2024*

- Conducted cohort and retention analysis using SQL and Looker Studio for merchant and user acquisition, revenue retention, and GMV retention to identify trends supporting strategic decisions to improve merchant engagement and business growth.
- Streamlined fraud detection by consolidating payment-gateway reports across gateways in SQL within Google BigQuery, reducing reporting time by 50%.
- Optimized and scaled daily auto-credit transfers for up to 2000 merchants daily, integrating the automation on Kubernetes using APIs.

Research Experience

Sample-Efficient Exploration in Online Reinforcement Learning — Sep 2025 – Present
New York University, Courant Institute | Prof. Rajesh Ranganath

- Developing ESAC, an ensemble RL method pairing independent diffusion/flow-matching actor-critic pairs with a Trajectory Integrated Disagreement Exploration (TIDE) bonus for sample-efficient online exploration.
- Outperformed strong baselines (SAC, REDQ) on MuJoCo locomotion and MetaWorld manipulation under matched compute; built the full training and evaluation stack end-to-end in JAX on a Slurm GPU cluster.
- Accepted for oral presentation at the ICML 2026 Demo Workshop (1 of 9 orals); under review at NeurIPS 2026.

Human-to-Robot Retargeting for Dexterous Manipulation — Sep 2025 – Present
New York University, General-Purpose Robotics and AI Lab | Prof. Lerrel Pinto

- Building a retargeting stack that maps human hand motion (vision keypoints) to a human-sized, tendon-driven humanoid hand (15-DoF, 11 actuators) via calibrated kinematics.
- Transforming human hand keypoints from video into temporally aligned robot trajectories via kinematic retargeting, integrating tactile feedback for slip detection and grip modulation to execute in-hand skills.
- Establishing evaluation on grasp taxonomy coverage and task success, with ablations on mapping strategy and coupling constraints; releasing reproducible configs, demonstration datasets, and benchmarks.

Visual Navigation using Semantic Conditional Constraint Feb 2024 – Oct 2024
University of Adelaide, Australian Institute of Machine Learning | *Prof. Sourav Garg*

- Predicted next actions from current visual observations and conditional constraints; used semantics for mapping with language input over current and goal images for context-aware planning.
- Experimented with SPOC, replacing navigation-camera RGB observations with category masks to move the robot through houses and reach language-specified goals.

Neural Integral Barrier Certificates Under Limited Actuation Feb 2024 – Sep 2024
University of Virginia, Chandra Robot Autonomy Lab | *Prof. Rohan Chandra*

- Developed a decentralized approach for safe, efficient, highly scalable multi-agent navigation guaranteeing collision-free movement and minimizing deadlock while satisfying input constraints.
- Achieved 100% collision avoidance and up to 92% deadlock reduction under limited actuation, scaling to 1024 agents (**IEEE ICRA 2025**).

Semi-Supervised Segmentation for PV Defect Detection Jan 2023 – Nov 2023
UCF, Center for Research in Computer Vision | *Prof. Yogesh Rawat, Prof. Shruti Vyas*

- Defect detection in photovoltaic cells via semantic segmentation of electroluminescence images using SegFormer and DeepLabv3+, with positive/negative thresholds in the loss to improve specific defects.
- Implemented a mean-teacher framework with only 20% labeled samples, achieving absolute gains of 9.7% IoU, 13.5% Precision, 29.15% Recall, and 20.42% F1 over prior SOTA supervised methods (*Engineering Applications of AI*, 2025).

Accepted Publications

- **Jha, A.**, Kar, S., Kumar, K., Parikh, M., Datta, S., Milani, S., Ranganath, R. (2026). ESAC: Ensemble Diffusion Actor-Critic with Trajectory-Integrated Disagreement Exploration for Online RL. **ICML 2026 Demo Workshop (Oral)**; under review at **NeurIPS 2026**. [paper]
- Xu, Y., **Jha, A.**, Chen, Y., Shen, Y., Heacock, L. (2026). Revealing Mammographic Phenotypes in Deep Learning Breast Cancer Risk Models. **MIDL 2026** (Poster). [paper]
- Zinage, V., **Jha, A.**, Chandra, R., Bakolas, E. (2025). Decentralized Safe and Scalable Multi-Agent Control under Limited Actuation. **IEEE ICRA 2025**. [paper] [project]
- **Jha, A.**, Rawat, Y., Vyas, S. (2025). Advancing automatic photovoltaic defect detection using semi-supervised semantic segmentation of electroluminescence images. **Engineering Applications of AI**, 160, 111790. [paper]
- **Jha, A.**, Gupta, T., Rawat, S.S., Kumar, G. (2024). Strategic Pseudo-Goal Perturbation for Deadlock-Free Multi-Agent Navigation in Social Mini-Games. **IEEE ICCRE 2024**, 264–269. [paper]
- **Jha, A.**, Mehta, I., Khan, K., Katarya, R. (2024). Enhancing ASD Diagnosis with Contrastive and Non-Contrastive Models from Neuroimaging Data. **IEEE ICMNWC 2024**.
- **Jha, A.**, Kumar, B., Madan, A.K. (2022). Real-Time Analysis of Material Removal Rate and Surface Roughness for Turning of Al-6061 using ANN and GA. **IJRESM**, 5(3), 145–150.

Projects

- **Ensemble Diffusion Actor-Critic with Disagreement Exploration (ESAC)**: [Project Page]
- **Hierarchical VLM Navigation with Diffusion Policy (VLM-Nav)**: [Code]
- **Distributed Sim Farm for Large-Scale RL Rollouts and Evaluation**
- **Multi-Robot Navigation in Social Mini-Games: Definitions, Taxonomy, and Algorithms**: [Project Page]
- **Autonomous Navigation of TurtleBot using SLAM**: [Code]
- **UAV Obstacle Avoidance using LiDAR**: [Code]

Skills

- **Languages**: C/C++, Python, R, MATLAB, Java, JavaScript, Go, LaTeX
- **Technologies**: Git/GitHub, MySQL, Linux, Spring Boot, Kubernetes, Slurm
- **Frameworks**: PyTorch, TensorFlow, Keras, CUDA, FastAPI, JAX, ROS, Rviz, Gazebo, Isaac Gym, PyBullet, MuJoCo
- **Others**: Data Structures, Probability & Statistics, Algorithms, NLP, Computer Vision, Machine Learning, Deep Learning, Reinforcement Learning, Data Mining, Regression Analysis